CHANDIGARH UNIVERSITY

C PROGRAMMING PROJECT REPORT

STUDENT MARKS MANAGEMENT SYSTEM USING ARRAY IN C PROGRAMMING

Submitted By	Lakshay Bihana
UID	26BCD10079
Section	26BCD2-A
Submitted To	Preeti
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Submitted in partial fulfilment of the C Programming project work.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my teacher, Preeti, for providing me with the opportunity to develop this C Programming project titled “Student marks management system Using C.” This project helped me understand how the concepts learned in class can be applied to create a working program.

I am thankful for the guidance and support that helped me organize the project and understand the logic behind displaying questions, checking answers, and calculating the final score.

I would also like to thank Chandigarh University for providing a learning environment that encourages practical knowledge and programming skills. This project has been a useful learning experience and has improved my confidence in writing and testing C programs.

1. INTRODUCTION

The Student Marks Management System is a simple C programming project designed to store, manage, and display students' marks using arrays and functions. The project demonstrates fundamental programming concepts such as arrays, user-defined functions, loops, conditional statements, and input/output operations.

In this program, the user enters the marks of students in different subjects. The program stores the marks in an array and uses functions to calculate the total marks, percentage, and grade of each student.

This project helps beginners understand how arrays and functions work together to organize and process data efficiently.

2. AIM OF THE PROJECT

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The aim of this project is to develop a C program to store and manage students' marks using arrays and user-defined functions. The program calculates the total marks, percentage, and grade of each student and displays the results. This project aims to develop an understanding of arrays, functions, loops, and conditional statements in C programming while improving logical thinking and problem-solving skill's

Expected Result

The expected result of this project is to develop a C program that successfully stores students' marks using arrays and functions. The program should accept marks for five subjects, calculate the total marks and percentage, assign grades based on the percentage obtained, and display the results accurately. It should also validate the entered marks to ensure they are within the range of 0 to 100. This project will demonstrate the effective use of arrays and user-defined functions in C programming.

3. OBJECTIVES OF THE PROJECT

The main objectives of this project are:

1. To understand the use of arrays in C programming.
2. To learn how to create and call user-defined functions.
3. To store and manage student marks efficiently.
4. To calculate total marks and percentage.
5. To determine grades based on students' performance.
6. To improve logical thinking and problem-solving skills.
7. To understand the practical application of C programming concepts.

4. HARDWARE AND SOFTWARE REQUIREMENTS

A. Hardware Requirements

• Computer or laptop: Usedto write,compile, and execute the program.

• Processor: A basic processor capable of running a C compiler.

• RAM: At least 2 GB of RAM is sufficient for this basic program.

• Keyboard: Used to enter answers and interact with the application.

B. Software Requirements

• OperatingSystem:Windows,Linux,or another operating system that supports a C compiler.

• Programming Language: C.

• Code Editor: Visual Studio Code (VS Code) for writing and editing the source code.

• C Compiler: GCC or another compatible C compiler to compile and execute the program.

These requirements are sufficient for developing and running this console-based application.

5. C PROGRAMMING CONCEPTS USED

1. Variables

Variablesstore information such as the user's selected answer and total score. In this project, integer variables are used for answer input and score calculation.

2. Functions and main()

Themain()functionis the startingpoint of the program. The program statements are arranged inside main() to control the quiz from beginning to end.

3. Conditional Statements

The ifstatementchecks whether theselected answer matches the correct option. If it matches, the score increases by one. The if-else-if structure is also used to display a performance message.

4. Input and Output

Theprintf()function displays questions, options, and results. The scanf() function accepts the user's answer from the keyboard.

5.Loops

1. For Loop: The for loop is used to input marks for multiple students and five subjects. It is also used to display student details, marks, total, percentage, and grade.
2. Do-While Loop: The do-while loop is used to validate marks entered by the user. If marks are below 0 or above 100, the program asks the user to enter them again until valid marks are provided.

6. ALGORITHM

The algorithm describes the logical sequence followed by the Student marks management system

Step 1:Start the program.

Step 2:Declare arrays to store student names and marks.

Step 3:Ask the user to enter the number of students.

Step 4:Accept each student's name and marks in five subjects.

Step 5:Validate that every mark is between 0 and 100.

Step 6:Calculate the total marks using a function.

Step 7:Calculate the percentage using a function.

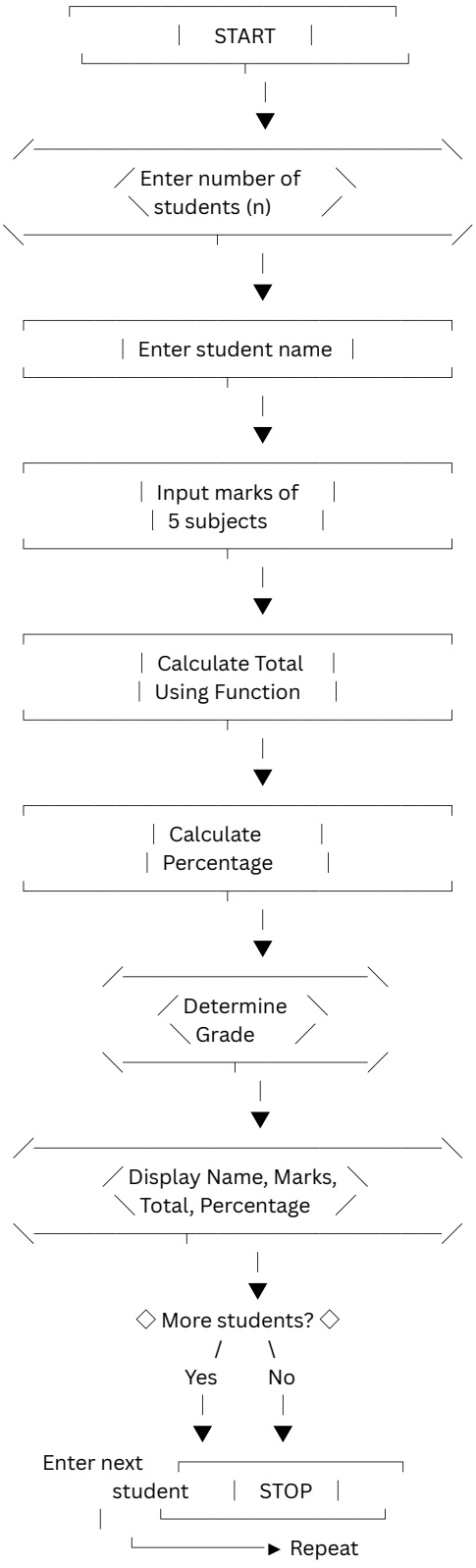
Step 8:Determine the grade using a function.

Step 9:Display each student's name, subject marks, total, percentage, and grade.

Step 10:Stop the program.

7. FLOWCHART

the flowchart for your **Student Marks Management System** based on the program you developed.
It shows the process from starting the program and entering student marks to calculating totals, percentages, grades, and displaying the final report.



8. SOURCE CODE

```
#include <stdio.h>
int main()
{
   char name[50];
   int english, computer, mathematics, physics, chemistry;
   int total;
   float percentage;
   printf("==================================\n");
   printf("   STUDENT MARKS MANAGEMENT SYSTEM\n");
   printf("==================================\n");
   printf("\nEnter student name: ");
   scanf("%49s", name);
   printf("\nEnter marks in English (out of 100): ");
   scanf("%d", &english);
   printf("Enter marks in Computer (out of 100): ");
   scanf("%d", &computer);
   printf("Enter marks in Mathematics (out of 100): ");
   scanf("%d", &mathematics);
   printf("Enter marks in Physics (out of 100): ");
   scanf("%d", &physics);
   printf("Enter marks in Chemistry (out of 100): ");
   scanf("%d", &chemistry);
   total = english + computer + mathematics + physics + chemistry;
   percentage = (total / 500.0) * 100;
   printf("\n==================================\n");
   printf("         STUDENT RESULT\n");
   printf("==================================\n");
   printf("Student Name: %s\n", name);
   printf("Total Marks: %d / 500\n", total);
   printf("Percentage: %.2f%%\n", percentage);
   if (percentage >= 90)
      printf("Grade: A+\n");
   else if (percentage >= 80)
      printf("Grade: A\n");
   else if (percentage >= 70)
      printf("Grade: B\n");
   else if (percentage >= 60)
      printf("Grade: C\n");
   else if (percentage >= 40)
      printf("Grade: D\n");
   else
      printf("Grade: F (Fail)\n");
         printf("==================================\n");
   return 0;
}
```

9. SAMPLE OUTPUT

The following is an example of how the program may appear when it is executed.

```
====================================
      STUDENT MARKS MANAGEMENT
                SYSTEM
====================================

  Enter Student Name: Lakshay Bihana
      Enter marks in Subject 1: 85
      Enter marks in Subject 2: 90
      Enter marks in Subject 3: 78
      Enter marks in Subject 4: 88
      Enter marks in Subject 5: 89

      ------------------------------------
              STUDENT RESULT
      ------------------------------------

      Student Name: Lakshay Bihana

          Subject 1 Marks: 85
          Subject 2 Marks: 90
          Subject 3 Marks: 78
          Subject 4 Marks: 88
          Subject 5 Marks: 89

        Total Marks: 430 / 500
          Percentage: 86%
          Grade: A

      ------------------------------------
       Result Generated Successfully
      ------------------------------------
```

Note: This is illustrative sample output. The score and performance message will change depending on the answers entered by the user.

10. CONCLUSION

The Student Marks Management System demonstrates the practical use of arrays and functions in C programming. It stores the marks of multiple students, calculates their total marks and percentages, assigns grades, and displays the results clearly.

Through this project, the concepts of arrays, loops, conditional statements, functions, and input validation can be understood more effectively. The project also provides a foundation for developing more advanced data management applications in C.

11. LEARNING OUTCOMES

After completing this project, I learned:

• How to write, compile, and execute a C program.

• How to declare and initialize variables.

• How to accept user input using scanf().

• How to display questions and results using printf().

• How to use if and if-else-if statements for decision-making.

• How to compare answers and update a score.

• How to test a program and verify its output.

• How programming logic can be used to create a simple interactive application.

This project strengthened my understanding of basic C programming and provided practical experience in developing a console-based application.